

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-211***

McConnellsburg 138 kV

(Revised)

October 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Community Energy Solar Development, LLC (“Interconnection Customer”) has proposed a new solar generation facility located in Fulton County, PA. The installed facilities will have a total capability of 70.0 MW with 48.1 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project is June 30, 2018. **This study does not imply a West Penn Power (“Transmission Owner”) commitment to this in-service date.**

Point of Interconnection

AC1-211 will interconnect with the West Penn Power transmission system by direct power injection into McConnellsburg substation. Please refer to the single-line diagram in Appendix 2 for system configuration.

Costs Summary and Transmission Owner Scope of Work

Transmission Owner facilities and network upgrades required to support AC1-211 project are as follows:

(a) Attachment Facilities: \$ 0.00

(b) Direct Connection Network Upgrades: \$ 0.00

McConnellsburg: Install New 138 kV Line Terminal, install 138 kV Breaker, Expand Substation Fence, Install Dual SEL421 Relays on New Terminal. Network Upgrade Number: to be determined at the facilities study stage.

Estimated cost: **\$ 1,164,900**

(c) Non-Direct Connection Network Upgrades: \$ 0.00

(d) Direct Connection Local Upgrades: \$ 0.00

(e) Non-Direct Connection Local Upgrades: \$ 0.00

(f) Option to Build Upgrades: \$ 0.00

Estimated Total Costs (a) to (f): \$ 1,164,900

NOTE: Above amounts are net and in 2017 Dollars. The Federal Income Tax Gross Up charge may or may not be added to above amounts depending on whether this project meets the eligibility requirements of IRS Safe Harbor Provisions for Non-Taxable Status.

Interconnection Customer Requirements

Interconnection Customer will be responsible for meeting all criteria as specified in the applicable sections of FirstEnergy “Requirements for Transmission Connected Facilities” document which posted on PJM website at <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>, including:

1. The purchase and installation of a fully rated circuit breaker on the high side of the AC1-211 step-up transformer.
2. The purchase and installation of the minimum required FirstEnergy generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of a 138 kV interconnection metering instrument transformer. FirstEnergy will provide the ratio and accuracy specifications based on the customer load and generation levels.
4. The purchase and installation of a revenue class meter for each unit to measure the power delivered in compliance with the FirstEnergy standards.
5. The purchase and installation of supervisory control and data acquisition (SCADA) equipment to provide information in a compatible format to the FirstEnergy Transmission System Control Center. The RTU, the communications channel and all related equipment will be furnished and maintained by Community Energy Solar Development, LLC. The RTU must communicate with the FirstEnergy EMS via DNP 3.0 protocol.
6. The establishment of dedicated communication circuits for SCADA report to the FirstEnergy Transmission System Control Center.
7. Interconnection Customer will be required to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the generator’s terminals.
8. The execution of a back-up service agreement, if required, to serve the Interconnection Customer’s load supplied from the AC1-211 POI when the units are out-of-service.
9. System Protection Analysis identified the following: Interconnection Customer to install at McConnellsburg Substation (1) Install fiber communication between FirstEnergy and the Interconnection Substation, and (2) Install Dual SEL421 over fiber communication with SEL501 BFT, (3) Install New 138 kV Terminal, and (4) Install New 138 kV Breaker
10. Interconnection Customer must also meet all PJM, ReliabilityFirst and NERC reliability criteria and operating procedures required for standards compliance. For example, the Interconnection Customer will need to properly locate and report the over and under-voltage and over and under-frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FirstEnergy system.
11. Compliance Issues: The proposed interconnection facilities must be designed in accordance with the FirstEnergy “Requirements for Transmission Connected Facilities” located at above shown link. Community Energy will also be responsible for following the requirements of the FirstEnergy “Approved Vendors and Contractors” document which is also located at the above link.

The above requirements are in addition to any metering required by PJM.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

FirstEnergy Requirements

The Interconnection Customer will be required to comply with all FirstEnergy Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "FirstEnergy Requirements for Transmission Connected Facilities" document located at the following links:

<http://www.firstenergycorp.com/feconnect>

<http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>

Schedule

Based on the scope of the attachment facilities, direct and non-direct system upgrades, it is expected to take a minimum of twelve (12) months from the date of a fully executed Interconnection Construction Service Agreement to complete the installation. This includes a full payment for the work that is related to this interconnection project. It also assumes that the Interconnection Customer will provide the property for the Project direct connection facilities and all right-of-way, permits, easements, etc. that will be needed. A further assumption is that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined direct connection facilities and that transmission system outages will be possible when requested.

Network Impacts

The Queue Project AC1-211 was evaluated as a 70.0 MW (Capacity 48.1 MW) injection into the McConnellsburg 138 kV substation in the APS area. Project AC1-211 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-211 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

None

Affected System Analysis & Mitigation

NYISO Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2020

Light Load Studies to be conducted during later study phases (as required by PJM Manual 14B).

System Reinforcements

Short Circuit

None

Stability and Reactive Power Requirement

To be determined at the facilities study stage.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Appendix 1

Facility Location

PJM Queue Position: AC1-211



Appendix 2

Interconnection One-Line Diagram

PJM Queue Position: AC1-211

